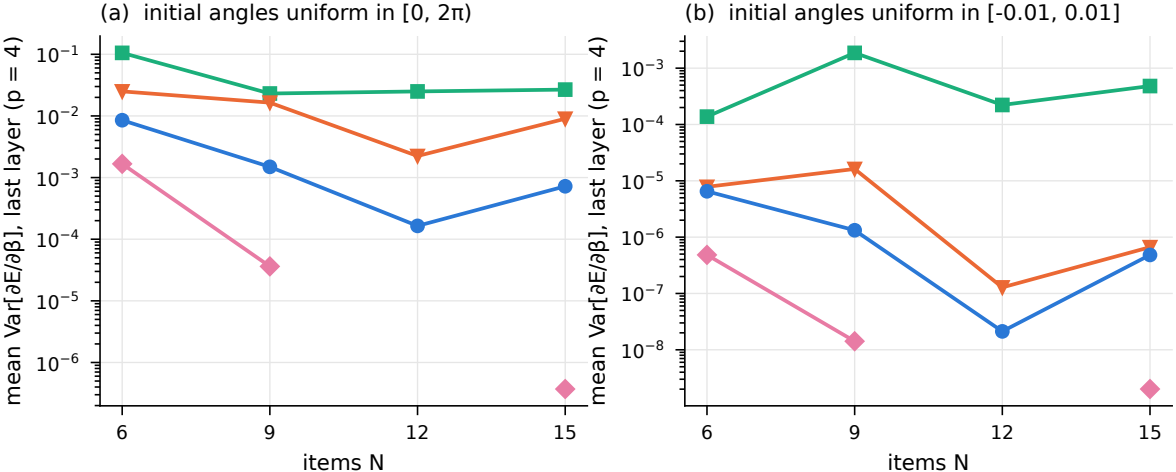
(a) initial angles uniform in [0, 2π)
(b) initial angles uniform in [-0.01, 0.01]
mean Var[∂E/∂β], last layer (p = 4)
items N
6
9
12
15
M1 MPSX-QAOA (proposed, multi-angle)
M2 MPSX-QAOA (single-angle)
M3 QAOA + slack penalty
M5 multi-angle particle-conserving (MA-PC)
gap = not simulated (memory cap): M5 at N = 12